E.G.S. PILLAY ARTS & SCIENCE COLLEGE
(AUTONOMOUS)
NAGAPATTINAM–611002. TAMILNADU
(Accredited by NAAC With "A" Grade)
(MBA, BBA & BCA–Approved by AICTE, New Delhi)
(Affiliated to Bharathidasan University,Tiruchirappalli–24)

B.Sc AI&ML- Bachelor of Science in Artificial Intelligence & Machine Learning

Syllabus

Choice Based Credit System-
Learning Outcomes Based Curriculum Framework
(CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

B.Sc AI&ML- Bachelor of Science in Artificial Intelligence & Machine Learning Choice Based Credit System Learning Outcomes Based Curriculum Framework (CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
I	Language Course – I (LC) –Tamil Other Languages	Tamil-I	6	3	3	25	75	100
	English Language Course -I(ELC)	English-I	6	3	3	25	75	100
	Core Course–I(CC)	Programming in C and Data Structures	6	4	3	25	75	100
	Core Practical-I(CP)	Programming in C Lab	4	4	3	50	50	100
	First Allied Course–I(AC)	Algebra & Calculus	6	4	3	25	75	100
	Value Education	Value Education	2	2	3	25	75	100
	Total		**30**	**20**				**600**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
II	Language Course – II (LC) -Tamil/Other Languages	Tamil-II	6	3	3	25	75	100
	English Language Course – II(ELC)	English-II	6	3	3	25	75	100
	Core Course–II(CC)	Programming in Python	5	4	3	25	75	100
	Core Practical-II(CP)	Programming in Python Lab	4	4	3	50	50	100
	First Allied Course–II(AC)	Numerical Analysis & Probability	3	2	3	25	75	100
	First Allied Course–III(AC)	Operation Research	4	3	3	25	75	100
	Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total		**30**	**21**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
III	Language Course – III (LC) -Tamil/Other Languages	Tamil-III	6	3	3	25	75	100
	English Language Course -III (ELC)	English-III	6	3	3	25	75	100
	Core Course – III (CC)	RDBMS and NoSQL	6	5	3	25	75	100
	Core Practical - III (CP)	RDBMS and NoSQL Lab	4	4	3	50	50	100
	Second Allied Course – I (AC)	Digital Computer Fundamentals	6	4	3	25	75	100
	Non Major Elective I	**A.** Fundamentals of Information Technology **B.** Basics of office automation	2	2	3	25	75	100
	Activity Based Course	Health & Wellness	-	1	-	-	-	100
	Total		**30**	**22**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
IV	Language Course –IV (LC) - Tamil/Other Languages	Tamil-IV	6	3	3	25	75	100
	English Language Course– IV (ELC)	English-IV	6	3	3	25	75	100
	Core Course–IV(CC)	Artificial Intelligence	5	4	3	25	75	100
	Core Practical-IV(CP)	Artificial Intelligence Lab	4	4	3	50	50	100
	Second Allied Course-II(AC)	Computer Organization and Architecture	4	3	3	25	75	100
	Second Allied Practical (AP)	Digital Computer Fundamentals Lab	3	2	3	50	50	100
	Non Major Elective II	A. Working Principles of Internet B. Web Designing	2	2	3	25	75	100
	Total		**30**	**21**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
V	Core Course V[CC]	Embedded Systems and IoT	5	5	3	25	75	100
	Core Course VI[CC]	Open Source Software	5	5	3	25	75	100
	Core Course VII[CC]	Robotics	6	5	3	25	75	100
	Core Practical V[CP]	Robotics Lab	5	4	3	50	50	100
	Major Based Elective-I	A. Virtual Reality and Augmented Reality B. Fuzzy Logic and Neural Networks	5	4	3	25	75	100
	Skill Based Elective- I	A. Mobile Application Development B. Social and Ethical Issues in Artificial Intelligence	2	2	3	25	75	100
	Internship	Internship (Report of the Study- 80 Marks, Viva Voce- 20 Marks)	-	2	-	20	80	100
	Soft Skills Development	Soft Skills Development	2	2	3	25	75	100
	Total		**30**	**29**				**800**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
VI	Core Course VIII[CC]	Human Computer Interaction	6	6	3	25	75	100
	Core Course IX [CC]	Machine Learning	6	6	3	25	75	100
	Core Practical VI[CP]	Machine Learning Lab	5	4	3	50	50	100
	Major Based Elective-II	A. Natural Language Processing B. Deep Learning	5	4	3	25	75	100
	Project	Project (Project Dissertation- 80 Marks, Viva Voce- 20 Marks)	5	3	3	20	80	100
	Skill Based Elective -II	A. Cloud Computing B. Cyber Security	2	2	3	25	75	100
	Extension Activities	NCC,NSS,YRC,YWF	-	1	-	-	-	-
	Gender Studies	Gender Studies	1	1	3	25	75	100
	Total		**30**	**27**				**700**
	Grand Total		**180**	**140**				**4200**

18/8/25

R. PRIYA 18.8.25
DR. R. PRIYA
ASSOCIATE PROFESSOR
SRMIST, CHENNAI

K. 2 18/8/25
Dr. K. UMAMAHESWARI
Assistant Professor.
AGCW, KARAIKAL

Dr. K. NANDHINI
Assistant Professor
Dept of Computer Science
Central University of Tamil Nadu, Thiruvarur

18/08/25
BOS - CHAIRMAN
Department of Artificial Intelligence & Machine Learning, Data Science and Computer Science with Cognitive Systems,
E.G.S.Pillay Arts & Science College (Autonomous),
Nagapattinam - 611 002.

B.Sc AI&ML- Bachelor of Science in Artificial Intelligence & Machine Learning Choice Based Credit System Learning Outcomes Based Curriculum Framework (CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
I	Language Course – I (LC) –Tamil Other Languages	Tamil-I	6	3	3	25	75	100
	English Language Course -I(ELC)	English-I	6	3	3	25	75	100
	Core Course–I(CC)	Programming in C and Data Structures	6	4	3	25	75	100
	Core Practical-I(CP)	Programming in C Lab	4	4	3	50	50	100
	First Allied Course–I(AC)	Algebra & Calculus	6	4	3	25	75	100
	Value Education	Value Education	2	2	3	25	75	100
	Total		**30**	**20**				**600**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
II	Language Course – II (LC) -Tamil/Other Languages	Tamil-II	6	3	3	25	75	100
	English Language Course – II(ELC)	English-II	6	3	3	25	75	100
	Core Course–II(CC)	Programming in Python	5	4	3	25	75	100
	Core Practical-II(CP)	Programming in Python Lab	4	4	3	50	50	100
	First Allied Course–II(AC)	Numerical Analysis & Probability	3	2	3	25	75	100
	First Allied Course–III(AC)	Operation Research	4	3	3	25	75	100
	Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total		**30**	**21**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
III	Language Course – III (LC) -Tamil/Other Languages	Tamil-III	6	3	3	25	75	100
	English Language Course -III (ELC)	English-III	6	3	3	25	75	100
	Core Course – III (CC)	RDBMS and NoSQL	6	5	3	25	75	100
	Core Practical - III (CP)	RDBMS and NoSQL Lab	4	4	3	50	50	100
	Second Allied Course – I (AC)	Digital Computer Fundamentals	6	4	3	25	75	100
	Non Major Elective I	**A.** Fundamentals of Information Technology **B.** Basics of office automation	2	2	3	25	75	100
	Activity Based Course	Health & Wellness	-	1	-	-	-	100
	Total		**30**	**22**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
IV	Language Course –IV (LC) - Tamil/Other Languages	Tamil-IV	6	3	3	25	75	100
	English Language Course– IV (ELC)	English-IV	6	3	3	25	75	100
	Core Course–IV(CC)	Artificial Intelligence	5	4	3	25	75	100
	Core Practical-IV(CP)	Artificial Intelligence Lab	4	4	3	50	50	100
	Second Allied Course-II(AC)	Computer Organization and Architecture	4	3	3	25	75	100
	Second Allied Practical (AP)	Digital Computer Fundamentals Lab	3	2	3	50	50	100
	Non Major Elective II	A. Working Principles of Internet B. Web Designing	2	2	3	25	75	100
	Total		**30**	**21**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
V	Core Course V[CC]	Embedded Systems and IoT	5	5	3	25	75	100
	Core Course VI[CC]	Open Source Software	5	5	3	25	75	100
	Core Course VII[CC]	Robotics	6	5	3	25	75	100
	Core Practical V[CP]	Robotics Lab	5	4	3	50	50	100
	Major Based Elective-I	A. Virtual Reality and Augmented Reality B. Fuzzy Logic and Neural Networks	5	4	3	25	75	100
	Skill Based Elective- I	A. Mobile Application Development B. Social and Ethical Issues in Artificial Intelligence	2	2	3	25	75	100
	Internship	Internship (Report of the Study- 80 Marks, Viva Voce- 20 Marks)	-	2	-	20	80	100
	Soft Skills Development	Soft Skills Development	2	2	3	25	75	100
	Total		**30**	**29**				**800**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
VI	Core Course VIII[CC]	Human Computer Interaction	6	6	3	25	75	100
	Core Course IX [CC]	Machine Learning	6	6	3	25	75	100
	Core Practical VI[CP]	Machine Learning Lab	5	4	3	50	50	100
	Major Based Elective-II	A. Natural Language Processing B. Deep Learning	5	4	3	25	75	100
	Project	Project (Project Dissertation- 80 Marks, Viva Voce- 20 Marks)	5	3	3	20	80	100
	Skill Based Elective -II	A. Cloud Computing B. Cyber Security	2	2	3	25	75	100
	Extension Activities	NCC,NSS,YRC,YWF	-	1	-	-	-	-
	Gender Studies	Gender Studies	1	1	3	25	75	100
	Total		**30**	**27**				**700**
	Grand Total		**180**	**140**				**4200**

Dr. K. NANDHINI
Asst Prof., Dept of Computer Science
Central University of Tamil Nadu
Thiruvarur.

Dr. K. Umamaheswari
Asst. Professor, Dept of CS
AGC&RI, Karaikal (Affiliated Pondicherry University)

Dr. R. PRIYA
Asst. Professor
Dept. of CSA
Faculty of Science & Humanities
SRMIST
Ramapuram, Chennai

BOS - CHAIRMAN
Department of Artificial Intelligence & Machine Learning, Data Science and Computer Science with Cognitive Systems,
E.G.S.Pillay Arts & Science College (Autonomous),
Nagapattinam - 611 002.

NON MAJOR ELECTIVE – I		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**III**	**A.** Fundamentals of Information Technology **B.** Basics of Office Automation

NON MAJOR ELECTIVE – II		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**IV**	**A.** Working principles of Internet **B.** Web Designing

SKILL BASED ELECTIVE–I		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**V**	**A.** Mobile Application Development **B.** Social and Ethical Issues in Artificial Intelligence

SKILL BASED ELECTIVE–I		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**VI**	**A.** Cloud Computing **B.** Cyber Security

LIST OF ALLIED COURSES

First Allied Course I — **Mathematics**

Second Allied Course II — **Applied Physics**

SUMMARY OF CURRICULUM STRUCTURE OF UG PROGRAMMES

Sl. No.	Part	Types of the Courses	No. of Courses	No. of Credits	Marks
1.	I	Language Courses	4	12	400
2.	II	English Courses	4	12	400
3.	III	Core Courses	9	44	900
4.		Core Practical	6	24	600
5.		Allied Courses I & II	5	16	500
6.		Allied Practical	1	2	100
7.		Major Based Elective Courses	2	8	200
8.		Internship	1	2	100
9.		Project	1	3	100
10.	IV	Non-Major Elective Courses	2	4	200
11.		Skill Based Elective Courses	2	4	200
12.		Soft Skills Development	1	2	100
13.		Value Education	1	2	100
14.		Environmental Studies	1	2	100
		Health & Wellness	1	1	100
15.	V	Gender Studies	1	1	100
16.		Extension Activities	1	1	--
	Total		**43**	**140**	**4200**

*for those who studied Tamil upto10th + 2 (Regular Stream) + Syllabus for other Languages should be on par with Tamil at degree level.

#those who studied Tamil up to 10th + 2 but opt for other languages in degree level under Part I should study special Tamil in Part IV.

**Extension Activities shall be outside instruction hours

** Internship are 10 to 15 days.

Note:

	Internal Marks	External Marks
1.Theory	25	75
2.Practical	50	50

1. Separate passing minimum is prescribed for Internal and External marks

FOR THEORY

The passing minimum for CIA shall be 40% out of 25 marks [i.e.10marks]

The passing minimum for University Examinations shall be 40% out of 75 marks [i.e.30 marks]

FOR PRACTICAL

The passing minimum for CIA shall be 40% out of 50 marks [i.e.20 marks]

The passing minimum for University Examinations shall be 40% out of 50 marks [i.e.20 marks]

PROGRAMME OUTCOMES (POs)

Upon completion of the undergraduate programme on B.Sc. Artificial Intelligence and Machine Learning, the students will be able to:

PO1: Apply the concepts and practical knowledge in analysis, design and development of computing systems and applications to multi-disciplinary problems.

PO2: Provide a concrete foundation and enrich their abilities to qualify for Employment, Higher studies and Research in Artificial Intelligence and Machine Learning with ethical values

PO3: Understand, analyze and develop essential proficiency in the areas related to artificial intelligence and machine learning in terms of underlying statistical and computational principles.

PO4: Learn the basic concepts of AI & ML and apply in various research areas like image processing, speech recognition and Medical diagnostics etc.

PO5: Find solutions to complex AI problems using various AI tools

PROGRAMME SPECIFIC OUTCOMES (PSOs)

After successful completion, the students were specially able to:

PO1: Apply the technical and critical thinking skills in the discipline of artificial intelligence and machine learning to find solutions for complex problems.

PO2: Design and create research – based solutions for challenging issues in the artificial Intelligence fields while giving due regard the issues of environmental, socioeconomic, cultural & public safety.

PO3: Establish the ability to listen, read, proficiently communicate and articulate complex ideas with respect to the needs and abilities of diverse audiences.

PO4: Exhibit good domain knowledge and completes the assigned responsibilities effectively and efficiently in expected quality standards for Artificial Intelligence and Machine Learning professional.

PO5: Provide innovative ideas to instigate new business ventures in the hospitality industry.

FIRST YEAR	CORE COURSE – I	COURSE CODE: 25AICC1
SEMESTER-I	PROGRAMMING IN C AND DATA STRUCTURES	CREDIT: 5

COURSE OBJECTIVES:

- To know about the basics of C Programming, Control and Looping Structure and programming with it.
- To understand Arrays, Pointers and String Processing in C language.
- To know about the basic concepts in Data Structures.

UNITS	CONTENTS	HOURS
I	Basic of C: History of C and its importance – Structure of a C program – Data Types – Constants and Variables – Operators and Expressions – Order of Precedence, Evaluating of Arithmetic Expressions – Type Conversion- Decision Statements: if, if-else, and nested if statements.	20
II	Loops Structures: For Loop, While, Do-while loop – Arrays: - One Dimensional Array, Two-dimensional Arrays, Character Arrays and Strings – Functions: Function with arrays- Function with decision and looping statements - Recursion.	20
III	Pointers: Introduction – Pointer Expressions – Chain of Pointers – Pointers and Arrays – Array of Pointers – Pointers as function arguments – Functions returning Pointers – Pointers to Functions – Function pointer – Structures - declaration, initialization, Array of Structures – Pointer to structures, Structures and functions – Typed of Enumerated data types, Unions.	20
IV	File Management in C: Introduction- Defining and Opening a File-Closing a File- Input/output Operations on Files-Error Handling During I/O Operations-Random Access to Files-Command Line Arguments.	15
V	Stack: LIFO concept, Stack operations, Array implementation of stack – Queue: FIFO concept, Queue operations, Array implementation of queue – Singly Linked List: concepts, operations – Doubly Linked List: concepts, operations – Trees : General trees, Binary trees.	15

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	To Summarize the basic knowledge to develop C programs	K2
CO2	To Manipulate Looping, arrays and functions	K4
CO3	To Apply and write programs for solving real world problems	K3, K2
CO4	To Create open, read, manipulate, write and close files.	K5
CO5	To Understand the basic concepts in data structures	K2

TEXTBOOKS:

1. E. Balagurusamy, "Programming in ANSI C", Tata McGraw Hill, New Delhi, Nineth Edition, 2024.
2. E. Horowitz, S. Sahni and Susan Anderson Freed, "Fundamental Data Structures in C", 2ed, Orient Black Swan Publisher, 2023.
3. Byron S. Gottfried, "Programming with C", Schaum's Outline Series, Tata- McGraw Hill Edition, New Delhi, 2025.

REFERENCES:

1. E. Karthikeyan, "A Textbook on C Fundamentals, Data Structures and Problem Solving", Prentice-Hall of India Private Limited, New Delhi, 2008.
2. Yashavant Kanetkar, "Let us C", BPB Publications, Tenth Edition, New Delhi, 2010.
3. Szuhay, Jeff, and Szuhay, Jeff, "Learn C Programming: A Beginner's Guide to Learning C Programming the Easy and Disciplined Way", Packt Publishing, 2020.
4. Jena, Sisir Kumar, and Jena, Sisir Kumar, "C Programming: Learn to Code", CRC Press, 2021.

WEB RESOURCES:

1. https://www.tutorialspoint.com/cprogramming/index.htm
2. https://www.w3schools.in/data-structures/intro

FIRST YEAR	CORE PRACTICAL - I	COURSE CODE: 25AICP1
SEMESTER-I	PROGRAMMING IN C LAB	CREDIT: 4

COURSE OBJECTIVES:

- To learn the programming knowledge using C language.
- To create derived and user defined data types.
- To demonstrate the functioning of pointers in C.

LIST OF PROGRAMS

1. Write a Program
    a. To convert temperature from degree Centigrade to Fahrenheit.
    b. To find whether the given number is Even or Odd.
    c. To find the greatest of three numbers.
2. Write a Program to use the switch statement to display Monday to Sunday.
3. Write a Program to display first Ten Natural Numbers and their sum.
4. Write a Program to find Multiplication of Two Matrices.
5. Write a Program
    a. To find the maximum number in Array using pointer.
    b. To reverse a number using pointer.
    c. To add two numbers using pointer.
6. Write a Program to solve Quadratic Equation using functions.
7. Write a Program to find factorial of a number using Recursion.
8. Write a Program to show Call by Value and Call by Reference.
9. Write a Program to create a file containing Student Details.
10. Write a program to implement a stack using singly linked list, Implement Queue using Linked List.

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able:

CO	COURSE OUTCOME	K LEVEL
CO1	To Relate the use of language constructs to solves impel programs	K4
CO2	To Develop programs for various concepts in C language	K4
CO3	To Understand and trace the execution of the list of programs	K2
CO4	To Understand the usage of file handling in C programming Solved at a problems related to data structures	K2
CO5	To Understand the basic concepts in C	K2

E.G.S.PILLAY ARTS & SCIENCE COLLEGE (AUTONOMOUS)
NAGAPATTINAM – 611 002. TAMIL NADU
(Accredited by NAAC With "A" Grade)
(MBA – Approved by AICTE, New Delhi)
(Affiliated to Bharathidasan University, Tiruchirappalli – 24)

ALLIED MATHEMATICS

for

B.C.A, B.Sc., Artificial Intelligence and Machine Learning, B.Sc., Computer Science, B.Sc., Information Technology Programmes

First Year	ALLIED COURSE - I			Credit: 4
Semester - I	ALGEBRA AND CALCULUS			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	6	-	-	6

COURSE OBJECTIVES:

- To lean the basic concepts in the integration.
- To train the students to solve the problems in Theory of Equations.
- To provide knowledge about the matrix, differentiation.

Units	CONTENT	HOURS
Unit I	**Theory of Equations:** Relation between roots & coefficients – Transformations of Equations – Diminishing, Increasing & multiplying the roots by a constant- Forming equations with the given roots.	18
Unit II	**Matrices:** Singular matrices – Inverse of a non-singular matrix using adjoint method - Rank of a Matrix – Consistency - Characteristic equation, Eigen values, Eigen vectors.	18
Unit III	**Differentiation:** Maxima & Minima – Concavity, Convexity – Points of inflexion - Partial differentiation – Euler's Theorem - Total differential coefficients (proof not needed) –Simple problems only.	18
Unit IV	**Integration :** Evaluation of integrals of types 1) $\int \frac{px+q}{ax^2+bx+c}dx$ 2) $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$ - Evaluation using Integration by parts.	18
Unit V	**Differential Equations:** Second order of types $(aD^2 + bD + c)y = F(x)$ where a, b, c are constants and $F(x)$ is one of the following types (i) e^{kx} (ii) $sin(kx)$ or $cos(kx)$ (iii) x^n, n being an integer (iv) e^{kx} $f(x)$.	18

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Train the students to solve the problems in theory of equations.	K1
CO2	Know about the origin of matrices and find the solutions of Eigen problems.	K2
CO3	Get exposed the basic concepts of differentiation and integration	K3
CO4	Acquire the knowledge about integral equations.	K4
CO5	Acquire the knowledge about differential equations.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	T.K. Manickavasagam Pillai & others, Algebra, Volume I, S.V Publications, 1985 Revised Edition (Units I, II).
2.	S. Narayanan, T.K. Manicavachagam Pillai, Calculus, (Volume II), S. Viswanathan Pvt Limited, 2003. (Units III, IV and V).
REFERENCE BOOKS	
1.	M.L. Khanna, Differential Calculus, Jaiprakashnath and Co., Meerut-2004
WEB RESOURCES	
1.	http://mathforum.org
2.	http://ocw.mit.edu/ocwweb/Mathematics

FIRST YEAR	CORE COURSE II	COURSE CODE: 25AICC2
SEMESTER-II	PROGRAMMING IN PYTHON	CREDIT: 5

COURSE OBJECTIVES:

- To develop programs using functions and pass arguments in Python.
- To write programs using loops and decision statements in Python.
- To design and program Python applications.

UNITS	CONTENTS	HOURS
I	Introduction to Python: Features of Python - How to Run Python - Identifiers - Reserved Keywords - Variables - Comments in Python - Indentation in Python - Multi-Line Statements - Multiple Statement Group (Suite) - Quotes in Python - Input, Output and Import Functions - Operators. Data Types and Operations: Numbers – Strings – List – Tuple – Set – Dictionary – Data type conversion.	15
II	Flow Control: Decision Making – Loops – Nested Loops – Types of Loops. Functions: Function Definition – Function Calling - Function Arguments - Recursive Functions - Function with more than one return value.	15
III	Modules and Packages: Built-in Modules - Creating Modules - import Statement - Locating Modules - Namespaces and Scope - The dir() function - The reload() function - Packages in Python - Date and Time Modules. File Handling- Directories in Python.	15
IV	Object-Oriented Programming: Class Definition - Creating Objects - Built- in Attribute Methods - Built-in Class Attributes- Destructors in Python – Encapsulation - Data Hiding – Inheritance - Method Overriding- Polymorphism.	15
V	Exception Handling: Built-in Exceptions-Handling Exceptions- Exception with Arguments - Raising Exception - User-defined Exception- Assertions in Python. Regular Expressions: The match() function - The search() function - Search and Replace - Regular Expression Modifiers: Option Flags-Regular Expression Patterns- Character Classes-Special Character Classes - Repetition Cases - findall() method - compile() method.	15

COURSE OUTCOMES:

Upon successful completion of this course the students would be able:

CO	COURSE OUTCOME	K LEVEL
CO1	To recall and understand the features of python programming language	K2
CO2	To illustrate various programming mechanism used in python	K2
CO3	To apply various language construct to write simple programs in python	K5
CO4	Write java To examine the application of object oriented concept in python programs using Applets	K2
CO5	To distinguish the various constructs used in python.	K5

TEXTBOOKS:

1. Jeeva Jose and P. Sojan Lal, "Introduction to Computing and Problem Solving with PYTHON", Khanna Book Publishing Co, 2016.
2. Mark Summerfield. — Programming in Python 3: A Complete introduction to the Python Language, Addison-Wesley Professional, 2018.
3. Satyanarayana, Radhika Mani, Jagadesh, "Python Programming", Universities Press(India) Pvt Ltd 2024.
4. Wesley J. Chun, "Core Python Programming", Prentice Hall Publication, 2006.

REFERENCES:

1. Timothy A Budd, "Exploring Python", Tata McGraw Hill, New Delhi, 2021
2. Jake Vander Plas, "Python Data Science Handbook: Essential Tools for Working with Data", O'Reilly Media, 2016.
3. Allen B. Downey, ``Think Python: How to Think Like a Computer Scientist, 2nd edition, Updated for Python 3, Shroff/O Reilly Publishers, 2016
4. Guido van Rossum and Fred L. Drake Jr, —An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd., 2021.

WEB RESOURCES:

1. https://www.w3schools.com/python/
2. https://docs.python.org/3/tutorial/index.html

FIRST YEAR	CORE PRACTICAL-II	COURSE CODE: 25AICP2
SEMESTER -II	PROGRAMMING IN PYTHON LAB	CREDIT: 4

COURSE OBJECTIVES:

- To develop the Numbers, Math functions and Strings.
- To create different Decision Making statements and Functions.
- To design GUI Applications in Python.

LIST OF PROGRAMS

1. Write a python program that displays the following information: Your name, Full address Mobile number, College name, Course subjects.
2. Write a python program to find the largest three integers using if-else and conditional operator.
3. Write a python program that asks the user to enter a series of positive numbers (The user should enter a negative number to signal the end of the series) and the program should display the numbers in order and their sum.
4. Write recursive functions for GCD of two integers.
5. Write recursive functions for the factorial of positive integer.
6. Write recursive functions for Fibonacci sequence up to given number n.
7. Write recursive functions to display prime number from 2 to n
8. Write a python program that writes a series of random numbers to a file from 1 to n and display.
9. Write a python program to sort a given sequence: String, List and Tuple.
10. Write a python program to make a simple calculator.

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able:

CO	COURSE OUTCOME	K LEVEL
CO1	To recall and relate the features of python programming language	K4
CO2	To compare various programming mechanism used in python	K4
CO3	To construct simple programs in python using various language features	K2
CO4	To distinguish the various constructs used in python	K2
CO5	To the application of object oriented concept in python	K2

First Year	ALLIED COURSE - II			Credit: 3
Semester - II	NUMERICAL ANALYSIS AND PROBABILITY			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	4	-	-	4

COURSE OBJECTIVES:

- To train the students in the numerical problems.
- To train the students in solving statistical problems.
- To learn knowledge about an algebraic and transcendental equations.

Units	CONTENT	HOURS
Unit I	**Algebraic & Transcendental equations:** Bisection Method, Newton Raphson Method, Iteration method - Finite differences –Forward, Backward differences.	12
Unit II	Numerical differentiation - Numerical Integration using Trapezoidal rule and Simpson's first & second rules (proof not needed) - Solutions to Linear Systems-Gauss elimination method.	12
Unit III	**Numerical solution of ODE:** Solution by Taylor Series Method, Euler's Method, Runge - Kutta 2nd order method.	12
Unit IV	Arithmetic Mean – Geometric Mean – Harmonic Mean - Median, Mode, Standard Deviation - Quartile Deviation.	12
Unit V	Correlation and Regression – Properties of Simple Correlation and regression coefficients – Simple Numerical Problems only.	12

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Solve algebraic and transcendental equations.	K1,K2
CO2	Understand the list of basic ideas of Numerical Methods and Statistics.	K2,K3
CO3	Solve differential and integral equations numerically.	K3
CO4	Analyze the accuracy and graphical representation of statistical data	K4
CO5	Understand the correlation and regression coefficients	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	S.S.Sastry, Numerical Analysis (Unit 1 , 2 , 3)
2.	Gupta.S.C & Kapoor, V.K, Fundamentals of Mathematical Statistics, Sultan Chand & sons, New Delhi -1994 (Units 4 & 5)
REFERENCE BOOKS	
1.	M.K. Jain, S.R.K. Iyengar and R.K. Jain, Numerical Methods for Scientific and Engineering Computation, New Age International Private Limited, 1999.

2.	C.E. Froberg, Introduction to Numerical Analysis, II Edn., Addison Wesley, 1979.
WEB RESOURCES	
1.	http://mathforum.org
2.	https://nptel.ac.in/courses/111/105/111105090/

First Year	ALLIED COURSE - III			Credit: 2
Semester - II	OPERATIONS RESEARCH			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	4	-	-	4

COURSE OBJECTIVES:

- To train the students to solve assignment problems, transportation problems.
- To train the students in network problems.
- To make students solve real life problems in Business and Management.

Units	CONTENT	HOURS
Unit I	**Operations Research:** Introduction - Basics of OR – OR & decision making – Role of Computers in OR - Linear programming formulations & graphical solution of two variables.	12
Unit II	Canonical & standard forms of LPP - Simplex Method : Simplex Method for < , = , > constraints – Charne's method of penalties.	12
Unit III	**Transportation problem:** Transportation algorithm –Degeneracy algorithm – Degeneracy in Transportation Problem , Unbalanced transportation problem- Assignment algorithm.	12
Unit IV	**Sequencing Problem:** Processing of n jobs through two machines – Processing of n jobs through 3 machines – processing of two jobs through m machines.	12
Unit V	**Networks:** Network – Fulkerson's rule - measure of activity – PERT computation – CPM computation.	12

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Recognize and relate LPP and solving LPP using graphical method.	K1
CO2	Compute Simplex Algorithm and Big-M Method of LPP.	K2
CO3	Explain Transportation problem and Evaluate its initial basic feasible solution and solve assignment problem using Hungarian algorithm.	K3
CO4	Sequencing the jobs to be carried out based on Cost Optimization.	K4
CO5	Describe and Construct Network and compute PERT and CPM.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	Manmohan & Gupta , Operations Research, Sultan Chand Publishers, New Delhi.
REFERENCE BOOKS	
1.	Prem Kumar Gupta and D.S. Hira, Operations Research : An Introduction, S. Chand and Co., Ltd. New Delhi.
2.	Hamdy A. Taha, Operations Research (7th Edn.), McMillan Publishing Company, New Delhi, 1982.
WEB RESOURCES	
1.	https://www.britannica.com/topic/operations-research
2.	http://ebooks.lpude.in.operationsresearch

SECOND YEAR	CORE COURSE III	COURES CODE: 25AICC3
SEMESTER -III	RDBMS AND NOSQL	CREDIT:5

COURSE OBJECTIVE:

- To know the basic concepts about database, its concepts, applications, data models, schemas and instances.
- To gain knowledge about database system architecture, the relational data model and about SQL.
- To Know about NoSQL and data models in NoSQL

UNIT	CONTENTS	HOURS
I	Introduction to Databases : Databases and Database Users - Introduction - Example - Characteristics of the Database Approach - Actors on the Scene - Workers behind the Scene - Advantages of Using the DBMS Approach - A Brief History of Database Applications.	15
II	Database System Concepts and Architecture : Data Models, Schemas, and Instances - Three-Schema Architecture and Data Independence - Database Languages and Interfaces - The Database System Environment - Centralized and Client/Server Architectures for DBMSs - Classification of Database Management Systems - The Relational Data Model and SQL : The Relational Data Model and Relational - Database Constraints - Relational Model Concepts - Relational Model Constraints and Relational Database Schemas - Update Operations, Transactions, and Dealing with Constraint Violations.	15
III	Basic SQL : SQL Data Definition and Data Types - Specifying Constraints in SQL -Basic Retrieval Queries in SQL - INSERT, DELETE, and UPDATE Statements in SQL - Additional Features of SQL - More SQL: Complex Queries, Triggers, Views, and Schema Modification - More Complex SQL Retrieval Queries - Specifying Constraints as Assertions and Actions as Triggers - Views (Virtual Tables) in SQL- Schema Change Statements in SQL.	15
IV	NoSQL : The Value of Relational Databases - Impedance Mismatch - Application and Integration Databases - Attack of the Clusters - The Emergence of NoSQL - Aggregate Data Models : Aggregates - Key-Value and Document Data	15

	Models - Column-Family Stores - Summarizing Aggregate-Oriented Databases.	
V	Details on Data Models: Relationships - Graph Databases - Schemaless Databases - Materialized Views - Modeling for Data Access - Distribution Models: Single Server – Sharding Master-Slave Replication - Peer-to-Peer Replication - Combining Sharding and Replication**.**	15

COURSE OUTCOMES:

Upon successful completion of this course the students would be able:

- To know about databases and about database users.
- To understand the concept of Database system structure and the concept of Relational model.
- To know about SQL.
- To know about the Emergence of NoSQL.
- To understand about data models in NoSQL.

Text Books

1. M. Morris Mano & Michael D. Ciletti, *Digital Design*, 6th Edition, Pearson Education, 2018.
2. Thomas L. Floyd, *Digital Fundamentals*, 11th Edition, Pearson Education, 2015.
3. Donald P. Leach, Albert Paul Malvino & Goutam Saha, *Digital Principles and Applications*, 8th Edition, McGraw Hill Education, 2015.

Reference Books

1. SQL: The Complete Reference, James R. Groff & Paul N. Weinberg, 3rd Edition, McGraw Hill Education, 2017.
2. Learning SQL, Alan Beaulieu, 3rd Edition, O'Reilly Media, 2020.
3. Seven Databases in Seven Weeks, Eric Redmond & Jim R. Wilson, 2nd Edition, Pragmatic Bookshelf, 2018.
4. MongoDB: The Definitive Guide, Kristina Chodorow, 3rd Edition, O'Reilly Media, 2019.

WEBSITE REFERENCES:

1. https://www.geeksforgeeks.org/dbms/
2. https://www.javatpoint.com/dbms-tutorial
3. https://www.tutorialspoint.com/dbms/index.htm
4. https://www.w3schools.com/sql/

SECOND YEAR	CORE PRACTICAL III	COURES CODE: 25AICP3
SEMESTER -III	RDBMS AND NOSQL LAB	CREDIT:4

COURSE OBJECTIVE:

- To practice the relational database functions using various operations
- To write queries in SQL to retrieve any type of information from a data base.
- To be able to understand unstructured table creation and processing using NoSQL.

Demonstrate the following SQL commands and can take any back end RDBMS system for implementation purpose.

EXERCISE	CONTENTS
1.	Write a SQL query for creating Table, and SQL queries for inserting, deleting, updating the records in Table.
2.	Write SQL Queries for AND/OR/NOT operation, Union-Intersection and Minus
3.	Write SQL queries for various Join Operations.
4.	Write SQL query for Sorting and Grouping the records.
5.	Write Nested queries, Sub queries using SQL.
6.	Write a SQL program using Built-in functions.
7.	Create a view and access the view using query.
8.	Creation of unstructured table contents using Nosql commands

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able:

- To work on database queries.
- To relate the entity relationship and join dependencies with software programs
- Write queries on aggregate functions, subqueires.
- Create structured and unstructured database using SQL and NoSQL
- Able to implement various functions of NoSQL.

WEBSITE REFERENCES:

1. https://www.oracle.com/database/technologies/appdev/sql.html
2. https://www.mongodb.com/docs/
3. https://neo4j.com/developer/graph-database/

SECOND YEAR	ALLIED COURSE-I	CREDIT :
SEMESTER - III	ALLIED PHYSICS-I (DIGITAL COMPUTER FUNDAMENTALS)	Subject code:

ALLIED PHYSICS-I
DIGITAL COMPUTER FUNDAMENTALS

Allied physics-I (only for AIML & CSCS students)

Objective :

To Understand the Number system, gates, Electronic circuits and Understand the Electronic combinational systems.

UNIT- I: Number system **(12 Hours)**

Number system – Binary, Octal, Decimal, Hexa decimal – Conversion – Binary codes – BCD code Gray – Binary Addition, Subtraction, Multiplication and Division.

UNIT- II: BOOLEAN ALGEBRA **(12 Hours)**

Boolean Algebra – Boolean Postulates and laws – De-Morgan's theorem – Simplication of Boolean expression – Karnaugh map.

UNIT- III: LOGIC GATES **(12 Hours)**

Logic gates – AND, OR, NOT, NAND, NOR and Ex-OR gates realization of gates using universal gates.

UNIT-IV: COMBINATIONAL CIRCUITS **(12 Hours)**

Arithmetic Circuit – BCD Addition, Subtraction, 1s complement and 2s complement – Half Adder and Full Adder – Half Subtractor and Full Subtractor.

UNIT – V: MEMORY DEVICES **(12 Hours)**

Classification of Memories –Primary Memory: RAM – ROM – Secondary Memory: hard Disk – Floppy Disk – Compact Disk – Digital Versatile Disk(DVD) – Flash Drive.

Total Hours- 60

Book for Study:

Digital Electronics-Meena. K

SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25AINM1A
SEMESTER - III	FUNDAMENTALS OF INFORMATION TECHNOLOGY	CREDIT: 2

COURSE OBJECTIVES:

- To familiarize the students with the world of IT and IT-enabled services.
- To provide an in-depth knowledge about internet and internet tools.
- To enable the students to understand about Computer Security.

UNIT	CONTENTS	HOURS
I	Introduction to Computers - Generation of Computers - Classification of Digital Computer - Anatomy of Digital Computer.	5
II	CPU and Memory - Secondary Story Devices - Input Devices - Output Devices.	5
III	Introduction to Computer Software - Programming Language - Operating Systems - Introduction to Database Management System.	5
IV	Computer Networks - WWW and Internet - Email - Web Design	5
V	Computers at Home, Education, Entertainment, Science, Medicine and Engineering - Introduction to Computer Security - Computer Viruses, Bombs, Worms.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand basic concepts and terminologies in IT and IT-enabled services.	K2
CO2	Understanding personal computers and their operations.	K2
CO3	Understand about operating systems and database management	K2
CO4	Comprehend about WWW, internet, email and web design concepts	K4
CO5	Respond to computer security issues.	K3

REFERENCES:

1. Alexis Leon and Mathews Leon, Fundamentals of Information Technology, Vikas Publishing House Pvt. Ltd, 2009
2. Fundamentals of Computers and Information Technology, M.N Doja,2005
3. Ramesh Bangia,"Computer Fundamentals and Information Technology", Laxmi Publications Pvt Limited, 2008.
4. Bharihoke, "Fundamentals of Information Technology", Excel Books, 2009. Ralph Stair, George Reynolds,
5. "Fundamentals of Information Systems" Cengage Learning, 2015.

SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25AINM1B
SEMESTER - III	BASICS OF AUTOMATION	CREDIT: 2

COURSE OBJECTIVES:

- To familiarize students with database concepts and data handling using MS-Access.
- To develop practical skills in using MS-Office applications for documentation & database management.
- To provide knowledge of spreadsheet operations, formulas, charts, and data analysis using MS-Excel.

UNIT	CONTENTS	HOURS
I	Introduction to MS – Windows, Operating System, Windows Desktop Icons and their functions, Dialog Boxes, Task Bar, Parts of Windows, Keyboard short keys or hot keys. Introduction to MS-Office.	5
II	MS-Word – creating, editing saving and printing text documents, Font formatting, inserting tables, smart arts, page breaks, working with images, Spelling and Grammar Check, document properties, Mail Merge.	5
III	MS-Excel – spreadsheet basics, creating, editing saving and printing spreadsheets, working with functions & formulas, modifying worksheets with color & auto formats, graphically representation : charts & graphs, data forms, analyzing data – securing spreadsheets	5
IV	MS-Power Point – creating, editing saving and printing slides, auto layouts, custom animation, slide transition, Graphical representation – charts & graphs, slide presentation.	5
V	MS-Access – planning a Database, access screen, creating database, tables, working with forms, creating queries, types of reports, printing & print preview, importing data.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand the basic concepts of operating systems, Windows environment, and MS-Office tools.	K2
CO2	Create, edit, format, and print professional documents using MS-Word.	K3
CO3	Prepare spreadsheets, apply formulas and functions, analyze data, and generate charts using MS-Excel.	K3
CO4	Create and manage simple databases, forms, queries, and reports using MS-Access.	K3
CO5	Apply office automation tools effectively for academic, personal, and professional purposes.	K3

REFERENCES:

1. Computer Fundamental & MS Office — Shashwat Publication, 2025 Edition. Covers Windows, MS Word, Excel, PowerPoint and Access aligned with NEP-2020 syllabus.
2. Technology For Success & The Shelly Cashman Series Microsoft 365 & Office Introductory — Cengage, 2025 Edition. Updated for Microsoft 365 and Office 2021 applications.
3. Keyboard 3: Windows 10 and MS Office 2016 — Oxford University Press India, 2025 Edition.

	COURSETITLE	YEAR	SEMESTER	CREDIT
	A.FOOD AND NUTRITIONAL BIOTECHNOLOGY	II	III	2

OBJECTIVES

- The students will be able to learn about food Preservation and microorganisms associated with itLearn about packaging of different foods. Learn types of Foods and their nutritional values
- Learn about components of foods and their deficiency disorders.5. Learn about Essential minerals , BMR and RDA

Unit I

Introduction to Food and Nutrition Biotechnology, Application of biotechnology to improve food quality, safety, production, and Nutritional value. Nutritional needs of the growing population.

Unit II

Food Fermentation Technology- Use of microorganisms to produce and preserve food products. Fermented foods - curd, cheese, bread, and vinegar have improved taste, shelf life, and nutritional value.

Unit III

Food Processing and Preservation- Food processing and preservation techniques help maintain food quality and prevent spoilage. Biotechnology contributes to the development of safer and longer-lasting food products through various preservation methods.

Unit IV

Nutritional Improvement of Foods Biotechnology is used to enhance the nutritional content of food through biofortification and genetic modification. Nutrient-enriched crops help reduce deficiencies and improve public health

Unit V

Functional foods and nutraceuticals - provide health benefits beyond basic nutrition, bioactive compounds that help prevent diseases and promote overall health and Well being.

REFERENCES

1. "Food Biotechnology "by Elsayed Abdel-Aaland And yKhatwa (2019)

2. "Introduction to Food Biotechnology" by Perry Johnson-Green(2016)

3. "Food Biotechnology "by Kalidas Shetty and Gopinadhan Paliyath (2005)

4. "Food Biotechnology, Second Edition "by Klaus Buchholz and Volker Kasche (2013)

5. "Nutritional Biochemistry and Metabolism: With Clinical Applications"by Maria Luz Fernandez and Jose M. Ordovas (2014)

6. "Biotechnology in Functional Foods and Nutraceuticals" edited by Debasis Bagchi and Francis, C. Lau (2010)

OUTCOMES

Understand food and nutrition biotechnology principles.

Explain microbial and enzymatic applications in food processing.

Apply biotechnology for food quality improvement.

Analyze nutritional and health benefits of foods.

Evaluate functional foods and nutraceuticals.

Understand food safety and modern biotechnological developments.

SECOND YEAR	CORE COURSE – IV	COURSE CODE: 25AICC4
SEMESTER - IV	ARITFICIAL INTELLIGENCE	CREDIT: 4

COURSE OBJECTIVES:

- To gain a basic idea of AI and AI related problem solving, inference, perception, knowledge representation and learning.
- To know about the Heuristic search techniques and about knowledge representation in AI.
- To know about Predicate Logic and about Reasoning in AI.

UNIT	CONTENTS	HOURS
I	Artificial Intelligence: AI Problems – Underlying Assumption – AI Technique – Level of the Model – Criteria of Success – Some General References. Problems, Problem Spaces, and Search: Defining the Problem as a State Space Search – Production Systems – Problem Characteristics – Production System Characteristics – Issues in the Design of Search Programs.	15
II	Heuristic Search Techniques: Generate and Test – Hill Climbing – Best-First Search – Problem Reduction – Constraint Satisfaction – Means-ends Analysis. Knowledge Representation Issues: Representations and Mappings – Approaches to Knowledge Representation – Issues in Knowledge Representation – The Frame Problem	15
III	Using Predicate Logic: Representing Simple Facts in Logic – Representing Instance and ISA Relationships – Computable Functions and Predicates – Resolution – Natural Deduction - Representating Knowledge Using Rules: Procedural Versus Declarative Knowledge – Logic Programming – Forward Versus Backward Reasoning – Matching – Control Knowledge	15
IV	Symbolic Reasoning Under Uncertainty: Introduction to Non-monotonic Reasoning – Logics for Non-monotonic Reasoning – Implementation Issues – Augmenting a Problem-solver – Implementation Depth First Search – Implementation Breadth First Search. Statistical Reasoning: Probability and Baye's Theorem – Certainty Factors and Rule- based Systems – Bayesian Networks – Dempster-Shafer Theory – Fuzzy Logic.	15
V	Semantic Nets- Frames.- Conceptual Dependency – Scripts – CYC. - Syntactic-Semantic Spectrum of Representation – Logic and Slot-and-Filler Structures – Other Representational Techniques.	15

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	To understand the basic idea of artificial intelligence and its application areas	K2
CO2	To apply basic principles of AI in solutions that requires problem solving.	K3
CO3	To demonstrate awareness and a fundamental understanding of various applications of AI techniques.	K2
CO4	To understand about Logic programming and about Reasoning related to AI.	K2
CO5	To know about the different representational techniques in AI.	K2

TEXTBOOKS:

1. Kevin Night, Elaine Rich, and Nair B., "Artificial Intelligence", McGraw Hill, 2017
2. M. Tim Jones, - Artificial Intelligence: A Systems Approach (Computer Science), Jones and Bartlett Publishers Inc.; First Edition, 2008.
3. Nils J. Nilsson, - The Quest for Artificial Intelligence, Cambridge University Press, 2009.

REFERENCES:

1. Gerhard Welss, - Multi Agents Systems, Second Edition, 2013
2. David L. Poole and Alan K. Mackworth, - Artificial Intelligence: Foundations of Computational Agents, Cambridge University Press, 2010.
3. Dan W. Patterson, "Introduction to AI and ES", PearsonEducation,2007.

WEB RESOURCES:

- https://www.ibm.com/topics/artificial-intelligence
- https://ai.stanford.edu
- https://www.csail.mit.edu/research/artificial-intelligence
- https://aitopics.org

SECOND YEAR	CORE COURSE – IV	COURSE CODE: 25AICP4
SEMESTER - IV	ARITFICIAL INTELLIGENCE LAB	CREDIT: 4

COURSE OBJECTIVES:

- To impart knowledge about the practical aspects in Artificial Intelligence related problems.
- To program different AI methods using a programming language.
- To know how the logical operations and reason based AI problems are using programming.

EXERCISE	CONTENTS
1.	Write a program to implement the Hill Climbing problem
2.	Write a program to implement the Towers of Hanoi problem
3.	Write a program to implement the Missionaries and Cannibals problem
4.	Write a program to implement the 8 queens problem
5.	Write a program to implement the A* Algorithm
6.	Write a program to Implement the Breadth first algorithm
7.	Write a program to implement the Depth first algorithm
8.	Write a program to implement the predicate logic

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Solve various kinds of problems using AI techniques	K3
CO2	Solve basic AI based problems using any programming language.	K3
CO3	Understand to implement the various kinds of AI based algorithms.	K3
CO4	Apply AI techniques to real-world problems to develop intelligent systems.	K4
CO5	To understand problems related to AI	K2

SECOND YEAR	ALLIED COURSE-II	CREDIT :
SEMESTER - III	ALLIED PHYSICS-II (COMPUTER ORGANIZATION AND ARCHITECTURE)	Subject code:

OBJECTIVES

- Features a shared memory for both data and programs, a processing unit (ALU)
- A CPU register that holds the memory address of the **next instruction** to be executed.
- Stores the instruction currently being executed by the CPU.
- Coordinates the sequence of data movements and execution within the processor.
- Hardwired control units use logic gates for fast execution, while microprogrammed units use software-like microinstructions stored in ROM

UNIT I: INTRODUCTION **(12 Hours)**

Functional units of digital system and their interconnections, buses, bus architecture, types of buses and bus arbitration. Register, bus and memory transfer. Processor organization, general registers organization, stack organization and addressing modes

UNIT II : ARITHMETIC AND LOGIC UNIT **(12 Hours)**

Look ahead carries adders. Multiplication: Signed operand multiplication, Booths algorithm and array multiplier. Division and logic operations.Floating point arithmetic operation, Arithmetic & logic unit design.

UNIT III: CONTROL UNIT **(12 Hours)**

Instruction types, formats, instruction cycles and sub cycles (fetch and execute etc), micro operations, execution of a complete instruction. Program Control, Reduced Instruction Set Computer, Pipelining. Hardwire and micro programmed control.

UNIT IV: MEMORY **(12 Hours)**

Basic concept and hierarchy, semiconductor RAM memories, 2D & 2 1/2D memory organization.ROM memories. Cache memories: concept and design issues & performance, address mapping and replacement. Auxiliary memories: magnetic disk, magnetic tape and optical disks. Virtual memory: concept implementation.

UNIT V: INPUT / OUTPUT **(12 Hours)**

Peripheral devices, I/O interface, I/O ports, Interrupts: interrupt hardware, types of interrupts and exceptions. Modes of Data Transfer: Programmed I/O, interrupt initiated I/O and Direct Memory Access., I/O channels and processors. Serial Communication: Synchronous & asynchronous communication, standard communication interfaces.

Total Hours- 60

REFERENCE BOOKS :

1. **Computer Organization and Design: The Hardware/Software Interface** by David A. Patterson & John L. Hennessy:
2. **Computer System Architecture** by M. Morris Mano
3. **Computer Architecture and Organization** by John P. Hayes
4. **Computer Organization and Architecture: Designing for Performance** by William Stallings

WEB SOURCES :

1. https://mrcet.com/downloads/digital_notes/ECE/III%20Year/07012023/COMPUTER%20ORGANIZATION%20AND%20ARCHITECTURE.pdf
2. http://home.ustc.edu.cn/~louwenqi/reference_books_tools/Computer%20Organization%20and%20Architecture%2010th%20-%20William%20Stallings.pdf
3. http://williamstallings.com/ComputerOrganization/

SECOND YEAR	ALLIED PRACTICAL COURSE-I	CREDIT :
SEMESTER - IV	ALLIED PHYSICS PRACTICAL -I (DIGITAL COMPUTER FUNDAMENTALS)	Subject code:

(only for AIML students)

Objective:

It promotes the exhaustive requirements and expectations of the students to acquire practical knowledge for the theory given in their syllabus.

1. Logic gates AND, OR, NOT.
2. NAND as an Universal gates
3. NOR as an Universal gates.
4. Demorgan's Theorem.
5. Half Adder and Full Adder.
6. Half Subtractor and Full Subtractor.
7. Boolean Expression
8. K- Map using logic gates.

Total Hours- 60

Book for Study:

Digital Electronics-Meena. K

TEXTBOOKS:

1."Computer Organization and Embedded Systems", 6th Edition by Carl Hamacher, McGraw Hill Higher Education.

2."Computer Organization and Design", The Hardware/Software Interface", 5th Edition by David A. Patterson and John L. Hennessy, Elsevier.

3.Computer System Architecture|, 3 rd Edition by M. Morris Mano, Pearson.

REFERENCES:

1. "Computer Architecture and Organization", 3rd Edition by John P. Hayes, WCB /McGraw-Hill

2. "Computer Organization and Architecture: Designing for Performance", 10th Edition by William Stallings, Pearson Education.

3. "Computer System Design and Architecture", 2nd Edition by Vincent P. Heuring and Harry F. Jordan, Pearson Education.

WEB RESOURCES:

- https://www.geeksforgeeks.org/computer-organization-and-architecture-tutorials/
- https://www.tutorialspoint.com/computer_logical_organization/index.htm
- https://www.javatpoint.com/computer-organization-and-architecture-tutorial
- https://www.ibm.com/topics/computer-architecture

22/5/26
Dr. K. KANDIHARI
Asst Prof. Dept of CS
Central University of Tamil Nadu
Thiruvarur.

22/05/2026
Dr. K. Umamaheswari
Asst. Prof. Dept of CS
AGCW, Karaikal.
[Affiliated Pondicherry University]

R. Priya/22.5.26
DR. R. PRIYA
ASST. PROF.
DEPT. OF LSA
Faculty of Science & Humanities
SRMIST
Ramapuram, Chennai

22/05/26
BOS - CHAIRMAN
Department of Artificial Intelligence & Machine Learning, Data Science and Computer Science with Cognitive Systems,
E.G.S.Pillay Arts & Science College (Autonomous),
Nagapattinam - 611 002.

SECOND YEAR	NON-MAJOR ELECTIVE- II	COURSE CODE: 25AINM2A
SEMESTER - IV	WORKING PRINCIPLES OF INTERNET	CREDIT: 2

COURSE OBJECTIVES:

- To teach the basics of the World Wide Web
- To understand the fundamentals of the Internet and the usage
- To know the components of Multimedia on the internet

UNITS	CONTENTS	HOURS
I	What is Internet? The Internet's underlying Architecture	5
II	Connecting to the Internet – Communicating on the Internet	5
III	How the World Wide Web works. Common Internet tools	5
IV	Multimedia on the Internet – Intranet and shopping on the Internet	5
V	Safeguarding the Internet	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand the evolution of the Internet.	K2
CO2	Know the basic knowledge of the web	K4
CO3	Comprehend the protocols and standards used throughout the Internet.	K2
CO4	Discuss a variety of Internet and WWW applications and related technologies.	K4
CO5	Evaluate the opportunities and threats created by interconnecting computers via the Internet.	K5

REFERENCES:

1. Preston Gralla, "How the Internet Works", Pearson Education, Eighth Edition, 2006.
2. C.Xavier, Fundamentals of Internet and Emerging Technologies, New Age International Private Limited; First Edition ,2021
3. Alexis Leon, Internet for Everyone, S. Chand (G/L) & Company Ltd; Second Edition 2012.
4. Andrea C. Nakaya,"Internet and Social Media Addiction", Reference Point Press,2015.
5. Richard Fox, Wei Hao,"Internet Infrastructure: Networking, Web Services, and Cloud Computing", CRC Press, 2017.

SECOND YEAR	NON MAJOR ELECTIVE II	COURSE CODE: 25AINMS2B
SEMESTER - IV	WEB DESIGNING	CREDIT: 2

COURSE OBJECTIVES:

- To understand the basic concepts of computer networks, client–server architecture, and web technologies.
- To gain knowledge of web browsing, search engines, email systems, and web navigation techniques.
- To design web pages using HTML including text formatting, lists, images, links, and tables.

UNIT	CONTENTS	HOURS
I	Introduction to Web related terms: Client –Server–Single Tier Architecture–Two Tier Architecture– N Tier Architecture – Internet – Intranet – WWW – Web Page – Web Site.	5
II	Introduction to Internet and its Protocols: Internet Addressing – IP Address – Domain Name – Electronic Mail–Uniform Resource Locator	5
III	Web Browser: Web Index – Web Search Engine–Meta Search Sites–Directory and Indexes – Managing Mails: Address Book – File Attachment Facility – Junk Mail	5
IV	Introduction to HTML: HTML Code for Web Page–HTML Tags-Web Page Basics–Display a Web Page in a Web Browser – Format texts – Font Style and sizes – Lists – Ordered Lists – Unordered Lists – Image – Add an image – Background Image – Horizontal Rule	5
V	Introduction to HTML Table: Cell Padding – Cell Spacing – Table Borders, HTML–Advantages and Disadvantages.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Explain the fundamentals of Internet, WWW, web architecture, and networking concepts.	K1
CO2	Identify and describe various Internet protocols and their roles in web communication.	K3
CO3	Use web browsers, search engines, and email systems effectively for information retrieval and communication.	K3
CO4	Design basic web pages using HTML elements such as text formatting, lists, images, and links.	K4
CO5	Create and manage HTML tables and apply formatting techniques for structured data representation.	K5

REFERENCES:

1. Alexis Leon & Mathews Leon, "Internet for Everyone", Leon Tech World, Chennai.2, 2009
2. John Zabour,JeffFoust & David Kerven,"HTML 4 HOW-TO",2005
3. Eric Kramer,"HTML". Kamalesh N.Agarwala, AmitLal & Deeksha Agarwala,"Business of the net, 2008.
4. Responsive Web Design with HTML5 and CSS (5th Edition) – *Ben Frain, Packt Publishing, 2025*

COURSE CODE	COURSE TITLE	YEAR	SEMESTER	CREDIT
	BIOTECHNOLOGY FOR HUMAN HEALTH	II	IV	2

Objectives:

- To develop new medicines, vaccines, and diagnostic methods for diseases
- To improve human health through genetic engineering and modern biotechnology techniques
- To prevent and treat infectious and genetic diseases effectively

Unit I

Industry Enzymes for textile industry, breweries, food supplements – single cell protein, vitamins, food processing - cheese, yoghurt making, biodegradable plastics, biofuels.

Unit II

Environment Applications of Biotechnology in environmental aspects: waste management, biodegradation of heavy metals, water cleaning, removing oil spills, air and soil pollution, bioremediation, biomining.

Unit III

Environment management - Environmental Impact Assessment - waste minimization techniques - environmental planning in urban development - natural resources and sustainable development - environmental ethics. tion, Increased milk production, artificial insemination, poultry and fisheries.

Unit IV

Applications in livestock improvement: transgenic animals, animal vaccine production .Transgenic crops for human

Unit V

Human Health and livestock Applications .Antibiotic production, Molecular diagnostics, vaccines and vaccine delivery, recombinant therapeutics – insulin, gene therapy, forensics.

Text Books/References

1. Bhasin, M.K. and Nath, S. 2002. Role of Forensic Science in the New Millennium, University of Delhi,

2. Crueger Wand Crueger, A. 2000. Biotechnology: A textbook of Industrial Microbiology. 2nd edition. Panima Publishing Co. New Delhi.

3. Eckert, W.G. and Wrightin, R.K. 1997. Introduction to Forensic Sciences, 2nd Edition, CRC Press, Boca Raton.

4. Hans-Joachim Jordening and Jesef Winter, 2005. Environmental Biotechnology Concepts and Applications.

5. James, S.H. and Nordby, J.J. 2005. Forensic Science: An Introduction to Scientific and Investigative Techniques, 2nd Edition, CRC Press, Boca Raton.